

Multiple Choice

1. **Answer:** A

Options: A) 275; B) 15; C) 20; D) 50; E) 11

Note: Correct option: A - 275

2. **Answer:** C

Options: A) $56\sqrt{15}$; B) $60\sqrt{7}$; C) $240\sqrt{7}/7$; D) $48\sqrt{15}$; E) $120\sqrt{15}$

Note: Correct option: C - $240\sqrt{7}/7$

3. **Answer:** A

Options: A) 125; B) 343; C) 64; D) 216; E) 512

Note: Correct option: A - 125

4. **Answer:** E

Options: A) 27 ft by 53 ft; B) 35 ft by 66 ft; C) 25 ft by 47 ft; D) 30 ft by 53 ft; E) 30 in. by 57 in.

Note: Correct option: E - 30 in. by 57 in.

5. **Answer:** A

Options: A) 0.7853981634; B) 1.0471975512; C) 1.8849555922; D) 0.6283185307; E) 0.2617993833

Note: Correct option: A - 0.7853981634

True / False

6. **Answer:** False

Options: A) True; B) False

Note: Actual answer is '-0.5'.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '-0.3926'.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'The population mean is also zero'.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'local maximum at P and a saddle point at Q'.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '0.1705'.

Long Form Response

11. **Answer:** 25.3 to 44.7 minutes. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** $\frac{\pi}{3}$. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key